NGUYỄN GIA NGHI

Nicholas Nguyen | Academic CV | Updated August 2026

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RESEARCH INTERESTS

Computer systems performance | Distributed systems & coordination | Software engineering

Latency diagnosis, Internet multimedia & dependable edge services

EDUCATION

VNUHCM-University of Science Sep 2023–May 2027

Bachelor of Science | Major: Information Technology Ho Chi Minh City

GPA: 8.34/10.00 (3.58/4.00) | 121 credits completed

Expected thesis completion: March 2027

UNDERGRADUATE RESEARCH THESIS

Latency diagnosis and optimization mechanisms for edge cloud-gaming systems | In progress

Supervisor: Professor Lê Ngọc Sơn | **Official title (Vietnamese):** Nghiên cứu cơ chế chẩn đoán và tối ưu hóa độ trễ cho hệ thống trò chơi đám mây biên

- Studying whether context-specific latency fingerprints - stored patterns of how the system responds to a bounded relief action - can help distinguish cross-layer bottlenecks and guide more targeted and accurate recovery than generic adaptation across heterogeneous edge-cloud-gaming environments.
- Built a working, self-hosted edge-cloud-gaming testbed with Dockerized game workloads, desktop host orchestration, GStreamer capture and encoding, WebRTC media and input paths, a lightweight browser client, and cross-layer telemetry.
- Implemented the first end-to-end fingerprinting slice, including versioned records, response-vector construction, fingerprint storage, context filtering, and evidence-backed match or “unknown” decisions. A held-out real run matched a compatible seed at 0.982 across 22 shared features, providing an early repeatability check.
- Planning controlled comparisons with fixed-profile, threshold-based and passive-telemetry baselines, using diagnosis accuracy, P95/P99 latency, frame-deadline misses, recovery time, quality degradation and probe overhead. A secondary comparison will test the controller with and without deadline-aware stale-work handling.

RELEVANT ACADEMIC PREPARATION

Selected strong results (10-point scale): Discrete Mathematics (9.40) | Introduction to Cloud Computing (9.10) | Computer Systems (9.10) | Software Design (8.50) | Applied Statistics for Engineers and Scientists (8.10)

Operating Systems (8.00) | Introduction to Databases (8.00)

Additional core coursework: Introduction to Software Engineering | Computer Networks | Data Structures and Algorithms

SELECTED SYSTEMS & SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine | Repository

- Developed a versioned Python package and CLI for schema validation, testbed-record ingestion, response construction, context-aware fingerprint matching, and conservative "unknown" results; the current automated suite contains 283 passing tests.

PIXELATED Studio Edition | Creator platform & self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into a self-hosted edge-cloud-gaming testbed that streams interactive workloads from a user-owned host to a lightweight browser client.
- Implemented a hosted Fastify control plane, Electron-based host orchestration, a Dockerized runtime, GStreamer capture and encoding, secure pairing and session flows, WebRTC media and input paths, and cross-layer telemetry.

Repository | Live application | Releases

PIXELATED User Edition | Browser-based WebAssembly gaming platform

- Built and deployed a React/PWA application that runs supported NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly-based libretro cores, without relying on the Studio desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser-execution baseline for comparison with Studio's edge-cloud-streaming path.

Repository | Live application

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern - TMA Solutions — Jun–Aug 2026
Ho Chi Minh City

- Worked with Jira tickets spanning bug fixes, product improvements, and small features across an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Modeled and queried MongoDB data through Mongoose, built API tests, self-tested changes, and actively participated in an Agile/Scrum workflow.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud platforms	AWS, GCP
Infrastructure & DevOps	Linux, Docker, Kubernetes, Helm, Terraform, GitHub Actions
Real-time media	WebRTC, GStreamer
Performance evaluation	Telemetry instrumentation, benchmarking, experimental design, latency analysis

CERTIFICATIONS & ENGLISH

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall - Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2 October 2025)